Shortest Distances in 3D: Planes

Questions 1–21

Question 1. Find the shortest distance from the origin to the plane with equation $x + 2y + 9z = 344$.

Question 2. Find the shortest distance between the point $A(2, 1, 2)$ and the plane with equation $x + y + z = 3$.

Question 3. Find the shortest distance from the given point P and the given plane Π:

a) $P(10, 3, 1)$, $\Pi : x - y - z = 3$

b) $P(-3, 2, 8)$, $\Pi : 2x + 5y - 8z = 6$

c) $P(-9, 0, 60)$, $\Pi : -5x - 3y + 9z = 10$

d) $P(6, 6, -1)$, $\Pi : -y - 2z = 1$

Question 4. It is given that the point A has position vector $-10\mathbf{i} + 20\mathbf{j} - 50\mathbf{k}$. Find the exact distance from the point A to the plane with equation $4x + y - 15z = 1698$.

Question 5. It is given that the lines l_1 and l_2 and the plane Π have equations:

$$l_1 : \mathbf{r} = \begin{pmatrix} 10 \\ 5 \\ -10 \end{pmatrix} + \lambda \begin{pmatrix} 30 \\ 1 \\ 5 \end{pmatrix}, \qquad l_2 : \mathbf{r} = \begin{pmatrix} 20 \\ 5 \\ 10 \end{pmatrix} + \mu \begin{pmatrix} -80 \\ -3 \\ 5 \end{pmatrix}, \qquad \Pi : x - 2y + 3z - 218 = 0.$$

The coordinates of where l_1 and l_2 intersect is point P.

a) Find the coordinates of point P.

b) Find the shortest distance of P from the plane Π.

Question 6. It is given that the plane Π has equation $3x - 2y - 14z + 39 = 0$ and that the line l has equation

$$\mathbf{r} = \begin{pmatrix} 5 \\ 28 \\ 18 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix}.$$

The point Q lies on the line l such that the shortest distance between Q and the plane Π is $\sqrt{209}$ units. Find the possible coordinates of Q.

Question 7. A large flat area of open ground is modelled as a plane with equation $z = 0$. A slope mine is dug due to a coal bed, which can be modelled as a plane with equation $\mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ 50 \end{pmatrix} = -4935$, where measurements are in metres. The entrance to the slope mine has coordinates $(25, 25, 0)$, relative to a fixed origin.

a) Find the shortest distance between the entrance to the slope mine and the coal bed.

Due to the shortest distance being a route that was too steep, it is instead decided to reach the coal bed via a tunnel that can be modelled by the equation

$$\mathbf{r} = \begin{pmatrix} 25 \\ 25 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 10 \\ 11/7 \\ 1 \end{pmatrix}.$$

b) Find the distance from the tunnel entrance to where the tunnel meets the coal bed.

Question 8. It is given that the points A, B and C have coordinates $(2, -1, 0)$, $(0, -2, -1)$ and $(2, 2, -2)$ respectively. It is also given that θ is the acute angle between the lines BA and BC. The triangle ABC lies in the plane Π.

a) Show that $\sin\theta = \frac{\sqrt{22}}{6}$.

b) Find the equation of the plane Π.

c) The point D has coordinates $(3, 1, 5)$. Show that D does not lie in the plane Π.

d) Find the shortest distance between D and the plane Π.

e) Hence find the exact volume of the tetrahedron $ABCD$, given that the volume is equal to $\frac{1}{3}\times$ area of base $\times$ perpendicular distance from base to apex.

Question 9.

a) Find the shortest distance from the origin to the plane Π with equation $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = 5$.

b) Show that the point A with coordinates $(-13, 4, 3)$ lies on the plane Π.

c) Write the equation of the plane Π in the form $\mathbf{r} \cdot \hat{\mathbf{n}} = d$, where $\hat{\mathbf{n}}$ is the unit vector in the direction of $\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$.

Question 10. Jake is trying to find the shortest distance from the point A with coordinates $(2, -3, 4)$ to the plane Π_1 with equation $\mathbf{r} \cdot \begin{pmatrix} -2 \\ 4 \\ 5 \end{pmatrix} = 8$. First, Jake finds the equation of Π_2, the plane that is parallel to Π_1 and contains A.

a) Find the equation of Π_2 in the form $\mathbf{r} \cdot \hat{\mathbf{n}} = d$.

b) Find Π_1 and Π_2 in the form $\mathbf{r} \cdot \hat{\mathbf{n}} = d$.

c) Show how Jake can use his two answers from part (b) to find the shortest distance from A to the plane Π_1.

Question 11. Find the shortest distance between the point $A(2, -10, -642)$ and the plane with equation $\mathbf{r} \cdot \begin{pmatrix} 3 \\ 9 \\ 1 \end{pmatrix} = 2$.

Question 12. Find the shortest distance from the origin to the plane with equation

$$\mathbf{r} = \begin{pmatrix} 10 \\ 66 \\ -8 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 5 \\ 7 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 3 \\ 9 \end{pmatrix}.$$

Question 13. Find the shortest distance from the point with coordinates $(2, 1, 7)$ to the plane with equation

$$\mathbf{r} = \begin{pmatrix} 120 \\ -15 \\ 985 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}.$$

Question 14. The plane Π contains both the point with coordinates $(2, 3, -5)$ and the line with equation

$$\mathbf{r} = \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}.$$

Find the shortest distance from the origin to the plane Π.

Question 15. Find the distance between the parallel planes $x + y + z = 40$ and $x + y + z = 60$.

Question 16. An artist is designing a sculpture made from several parallel pieces of acrylic. Two of the acrylic pieces can be modelled by the planes given by

$$\Pi_1 : px + 2y + qz = 300, \qquad \Pi_2 : px + 2y + qz = 627.$$

It is given that the acrylic piece modelled by Π_1 must pass through the points with coordinates $(100, -150, -4)$ and $(150, -100, 10)$.

a) Find the values of p and q.

b) Find the distance between the planes Π_1 and Π_2.

Question 17. The planes Π_1 and Π_2 are parallel. Given that Π_1 has equation $x + 2y + 2z = 10$ and that the distance between Π_1 and Π_2 is 5 units, find the two possible equations of plane Π_2.

Question 18. A laser is set up between two large parallel mirrors where distances are in metres. Given that $p > 0$, the mirrors can be modelled by the equations

$$\Pi_1 : px + 9y + 3z = 6, \qquad \Pi_2 : px + 9y + 3z = 26.$$

When fired, the laser passes through the points with coordinates $(4, 2, 0)$ and $(4, 10, 8)$. The laser first hits Π_1, making an acute angle θ with the normal to Π_1 such that $\sin\theta = \frac{\sqrt{6}}{3}$.

a) Find the value of p.
b) Find the coordinates of where the laser first hits Π_1.
c) Find the distance between the two mirrors.

Question 19. Find the distance between the line with equation

$$\mathbf{r} = \begin{pmatrix} 2 \\ 8 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix}$$

and the plane with equation $-31x + 14y + 13z + 2615 = 0$.

Question 20. Find the distance between the line with equation $2x = \frac{y-3}{2} = \frac{60-6z}{13}$ and the plane with equation

$$\mathbf{r} \cdot \begin{pmatrix} 3 \\ -4 \\ -3 \end{pmatrix} = 60.$$

Question 21. It is given that the point $P(2, 1, 2)$ is the same distance from the plane $4x + y + 8z = -65$ as it is from the line

$$\mathbf{r} = \begin{pmatrix} 0 \\ -13 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 3 \\ k \end{pmatrix}.$$

Find the value of k.